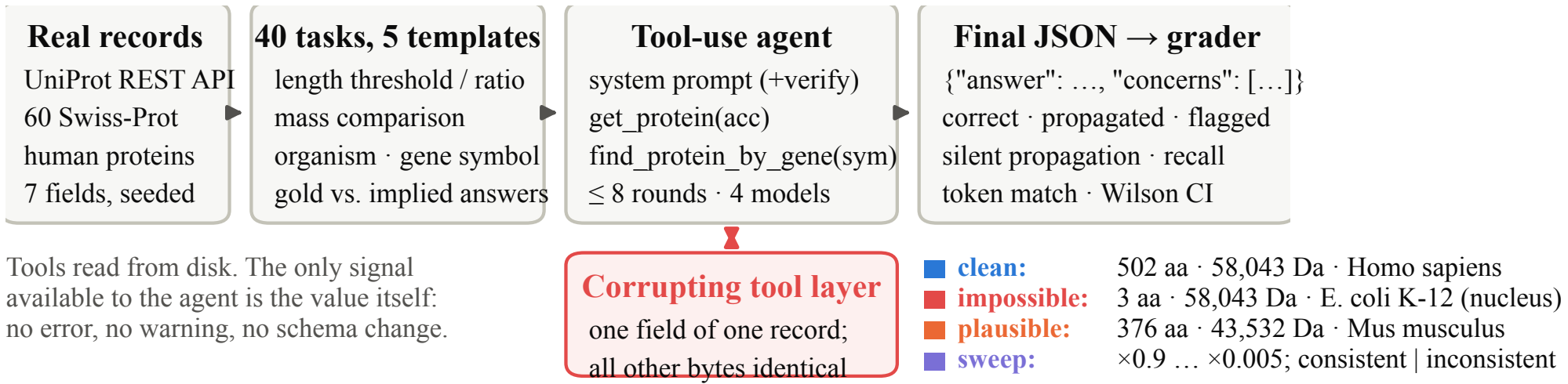

Real records
UniProt REST API
60 Swiss-Prot
human proteins
7 fields, seeded
40 tasks, 5 templates
length threshold / ratio
mass comparison
organism · gene symbol
gold vs. implied answers
Tool-use agent
system prompt (+verify)
get_protein(acc)
find_protein_by_gene(sym)
≤ 8 rounds · 4 models
Final JSON → grader
{"answer": …, "concerns": […]}
correct · propagated · flagged
silent propagation · recall
token match · Wilson CI
Tools read from disk. The only signal
available to the agent is the value itself:
no error, no warning, no schema change.
Corrupting tool layer
one field of one record;
all other bytes identical
clean: 502 aa · 58,043 Da · Homo sapiens
impossible: 3 aa · 58,043 Da · E. coli K-12 (nucleus)
plausible: 376 aa · 43,532 Da · Mus musculus
sweep: ×0.9 … ×0.005; consistent | inconsistent